SHRINCS specification

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Abstract

Oleks: To update after the spec is ready This document specifies **SHRINCS** (Stateful Hash-based Robust Integrated NIST-Compatible Signatures), a post-quantum signature scheme optimized for Bitcoin. SHRINCS combines efficient stateful hash-based signatures (unbalanced XMSS with WOTS+C) for normal operation with a stateless fallback (SPHINCS+ variant) for scenarios when state management failed. The scheme achieves minimal 324-byte signatures during normal stateful operation while maintaining the robustness of stateless signatures as a backup when state is corrupted or lost.

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1 Motivation

Current Bitcoin signature schemes (ECDSA[**ecdsa**] and Schnorr[**bip341**] over secp256k1) are vulnerable to quantum attacks via Shor's algorithm[**shor**]. Hash-based signature schemes offer post-quantum security relying only on the security of cryptographic hash functions, which are believed to resist quantum attacks apart from Grover's quadratic speedup (mitigated by doubling hash output lengths).

The key challenges for deploying hash-based signatures in Bitcoin are:

1. Signature size: NIST-standardized SLH-DSA signatures [**slh-dsa**] range from 7KB to 50KB, significantly larger than current 64-byte Schnorr signatures. This increases transaction fees and reduces the network's throughput.

2. Stateful vs. Stateless tradeoff: Stateful schemes (e.g., XMSS [**xmss**]) offer compact signatures but require careful state management to prevent key reuse. Additionally, Bitcoin users expect to restore wallets from static seed phrases (BIP-39[**bip-39**]). Pure stateful schemes break this model since signing state cannot be recovered from a seed alone, the signer must know which one-time keys have already been used. Stateless schemes (e.g., SLH-DSA) are robust but produce large signatures.

1.1 SHRINCS

SHRINCS (Stateful Hash-based Robust Integrated NIST-Compatible Signatures) is a hybrid post-quantum signature scheme that combines the efficiency of stateful signatures with the robustness of stateless ones:

- **Efficient stateful path.** During normal operation with intact state, SHRINCS uses an Unbalanced XMSS (UXMSS) tree with WOTS+C one-time signatures. Signature size is $308 + 16q$ bytes (where $q \geq 1$ is the signature index), yielding 324 bytes for the first signature, which is $\times 20$ smaller than the smallest SLH-DSA variant.

- **Robust stateless fallback.** When the state is lost, corrupted, or exhausted, SHRINCS falls back to a SPHINCS+-based stateless signature (3992 bytes). This is still smaller than standard SLH-DSA.

- **Static seed backup.** Users can back up their wallet using a standard seed phrase. Upon recovery, the wallet operates in stateless-only mode, preserving funds while accepting larger signatures.

- It provides with two parameters set depending on what matters for verifier:

  - SHRINCS-B (Bitcoin-optimized): Minimizes signature size at the cost of more verification hashes.
  - SHRINCS-L (Liquid-optimized): Minimizes verification cost at the cost of larger signatures.

1.2 Limitations and Practical Considerations

Users and developers should be aware of the following limitations inherent to stateful signature schemes:

State Management Across Devices. The signing state (specifically, the counter q indicating which one-time keys have been already used) cannot be safely shared across multiple devices without coordination. Two devices signing with the same q value would reuse a one-time signature, which leads to key compromise. Some mitigation strategies can be:

- Dedicate disjoint ranges of leaf indices to different devices.

- Use a single signing device with secure state storage (other devices are stateless backups).

- Accept stateless-only operation for multi-device wallets.

State Corruption and Power Failure. If a signing operation is interrupted (e.g., due to power failure) after producing a signature but before persisting the updated state, the signer may not know whether the one-time key was used. Subsequent signing attempts risk catastrophic key reuse. Approaches to mitigate:

- Incrementing the state counter before the signature. If signing fails, a leaf is wasted but security is preserved.
- State corruption detection, to maintain checksums or redundant state copies to detect corruption. If corruption is detected, fall back to stateless mode.[1]

State Growth with Multiple Addresses. Each SHRINCS-based address requires its own signing state. As users generate new addresses (as recommended for privacy), they must maintain state for each address that may be used for future spending. This state must be preserved for as long as the address holds funds. Mitigation:

- Wallet backups must include current state for all active addresses, not just the seed.
- Addresses can be marked as "stateless-only" after backup, accepting larger signatures for those addresses.

Variable Stateful Signature Sizes. SHRINCS signatures presume variable signature size and verification complexity depending on: (1) the number of already produced stateful signatures; (2) the signing mode. In decentralized system we need to consider how the cost of the signature verification is calculated to prevent spam attacks.

- That's basically the reason this specification introduced two modes for SHRINCS. SHRINCS-B is designed for accounting systems, where the user pays for each transaction byte, so the signature size matters. SHRINCS-L suits better for systems which calculate the transaction cost based on how many computation units it consumes (like Liquid[2]).

2 Notation

Throughout this specification:

`||` denotes byte string concatenation

`[a:b]` denotes a substring from byte index `a` to `(b-1)`

`[0:a]` denotes first `a` bytes (truncation)

`←$` uniform random sampling

`n` is the security parameter in bytes

`q` is the stateful signature counter

`H` denotes a cryptographic hash function

`Tw` denotes a tweakable hash function

`PRF` denotes a pseudorandom function

$\perp$ used for failure returns

[1] Here is an explicit guide for how to manage a state and backups for post-quantum constructions: https://datatracker.ietf.org/doc/draft-ietf-pquip-hbs-state/

[2] **Oleks:** Make a reference how WU are calculated for liquid and why signatures with bigger sizes can result to cheaper transactions

3 Overview

SHRINCS is designed in the way it can operate in two modes: **stateful** and **stateless**. Stateful mode is represented by the instance of Unbalanced XMSS (UXMSS) with a root public key `PK.sf`. An optimized variant of SPHINCS+ is used to support the stateless operating mode (which is represented by the public key `PK.sl`). The root public key is a hashed concatenation of (`PK.sf`, `PK.sl`), which is being converted to the Bitcoin address.

During normal operation with intact state, the user generates signatures according to efficient stateful path. Each leaf of the UXMSS is represented by WOTS+C one time signature instance. UXMSS is right-skewed, which makes the first stateful signature (`q=1`) the cheapest from the verification and size perspective (324 bytes) and adds the constant size (16 bytes) and one additional hashing operation for each next stateful signature. A right-skewed UXMSS is a tree construction in which every internal node has a one-time signature leaf as its left child and a UXMSS subtree as its right child, yielding a recursively right-branching authentication tree (as shown on the Figure 1).

If the state is corrupted (or the limit for stateful operations is met), the scheme can continue operating with stateless mode. Stateless mode utilizes an optimized SPHINCS+ variant with WOTS+C signatures in regular XMSS trees on each layer in the hypertree and FORS+C for few-time signature scheme. The general construction of SHRINCS signature is presented on the Figure 1.

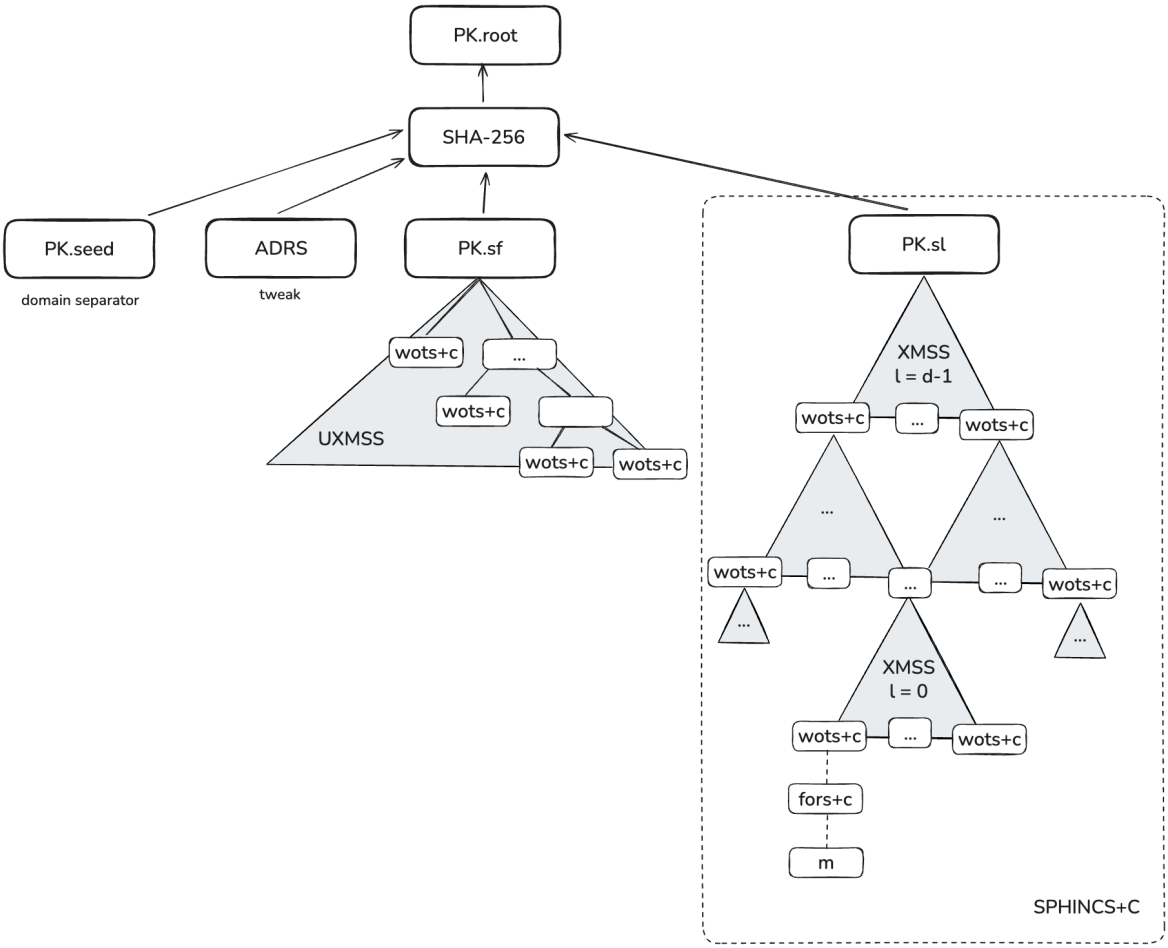

Figure 1: SHRINCS architecture

3.1 Key Pair Derivation and Size

SHRINCS keys are derived from a single random seed (**Oleks:** to add notes about BIP-39 usage and hd wallets if needed, maybe in appendix):

```
seed ←$ {0,1}^(3*8*n) // 3n bytes
```

Seed allows to derive a private key `SK ← (SK.seed, SK.prf, PK.seed, PK.sf, PK.sl, PK.root)`:

```
SK.seed ← seed[0:n]  // Seed for key derivation
SK.prf ← seed[n:2n]  // Secret needed for PRF_msg
PK.seed ← seed[2n:3n] // Domain separator for different SHRINCS key pairs
PK.sf ← compute_stateful_root(PK.seed, ADRS, SK.seed) // Public stateful key, see Sec.~10
PK.sl ← compute_stateless_root(PK.seed, ADRS, SK.seed) // Public stateless key, see Sec.~10
PK.root ← H(PK.seed, ADRS, PK.sf, PK.sl) // Root public key
```

where `ADRS` is a tweak also known as a context address (see Sec. 5.5).

The size of the secret key is `6n` bytes. A public key consists of two values `PK ← (PK.seed, PK.root)` and has a size of `2n` bytes.

3.2 Signature Size and Verification Cost

3.2.1 Stateful

Stateful signatures consist of:

1. `PK.sl`: the stateless public key (for unified verification)
2. `R`: randomness for message hashing (32-byte value)
3. `ctr`: grinding counter
4. `chain_values`: WOTS+C signature (`l` chains)
5. `auth_path`: UXMSS authentication path (q nodes)

Signature size calculation:

```
General formula: size = n  + R_SIZE + ctr_SIZE +  l * n +  q*n
       SHRINCS-B.size = 16 +    32  +    4     + 16 * 16 + q*16 = 308 + 16q
       SHRINCS-L.size = 16 +    32  +    4     + 64 * 16 + q*16 = 1076 + 16q
```

The signature verification cost is:

```
General formula: hash_ops = (w-1) * l -  S_wn + q + 1
       SHRINCS-B.hash_ops =  255  * 16 - 2040 + q + 1 = 2041 + q
       SHRINCS-L.hash_ops =   3   * 64 - 140  + q + 1 = 53 + q
```

3.2.2 Stateless

Stateless signatures consist of:

1. `PK.sf`: the stateful public key
2. FORS+C signature: few-time signature on the message
3. HT signature: `d` XMSS signatures forming the hypertree

Signature size calculation:

```
General formula: size  = fors_size(R_SIZE + ctr_SIZE + (k-1)*(n + a*n))
                       + xmss_layer_size(R_SIZE + ctr_SIZE + l*n + h'*n) * d
      SHRINCS-B. size  = fors_size(32 + 4 + 5*(16 + 22*16))
                       + xmss_layer_size(32 + 4 + 16*16 + 8*16) * 5
                       = 3992
      SHRINCS-L. size  = fors_size(32 + 4 + 5*(16 + 22*16))
                       + xmss_layer_size(32 + 4 + 64*16 + 8*16) * 5
                       = 7832
```

The signature verification cost is:

```
General formula:
    fors_hashes = (k-1)*(1 + a) + 1
    wots_per_layer = (w-1)*l - S_wn + 1
    xmss_per_layer = wots_per_layer + h'
    total = fors_hashes + d*xmss_per_layer

SHRINCS-B:
    fors_hashes = 5*23 + 1 = 116
    wots_per_layer = (256-1)*16 - 2040 + 1 = 2041
    xmss_per_layer = 2041 + 8 = 2049
    total = 116 + 5*2049 = 10361 hashes

SHRINCS-L:
    fors_hashes = 116
    wots_per_layer = (w-1)*l - S_wn + 1 = 53
    xmss_per_layer = wots_per_layer + h' = 53 + 8 = 61
    total = fors_hashes + d*xmss_per_layer
          = 116 + 5*61 = 421 hashes
```

Mike: I am not sure what is the difference between SHRINCS-B and SHRINCS-L. Are these two modes of opertaions? With different parameter sets. If so - I think we should add some more explanation to the text.

4 Parameters

This specification provides with the list of parameters for different SHRINCS instances, depending on verifier requirements:

1. SHRINCS-B: size matters (instance preferable for Bitcoin).
2. SHRINCS-L: verification complexity matters (the number of hashes to verify the signature).

Parameter	SHRINCS-B	SHRINCS-L	Description
`H`	`SHA-256`	`SHA-256`	Hash function
`OTS`	`WOTS+C`	`WOTS+C`	One-time signature
`FTS`	`FORS+C`	`FORS+C`	Few-time signature
`n`	16	16	Security parameter (bytes)
`w`	256	4	Winternitz parameter
`l`	16	64	The number of WOTS+C chains
`Swn`	2040	140	Target sum for WOTS+C
`hsf`	226	418	Maximum stateful tree height[3]
`hsl`	40	40	Maximum stateless hypertree height
`d`	5	5	Stateless hypertree layers
`h'`	8	8	XMSS tree height per hypertree layer `hsl/d = 8`
`a`	22	22	FORS tree height
`k`	6	6	Number of FORS+C trees
`c`	4	4	Counter size for WOTS+C grinding (bytes)

Mike: We can also explore the option of pushing Target Sum value higher (espessilay for smaller SHRINCS-L parameters to make verification even cheaper)

5 Underlying Functions and Addresses

This section defines the cryptographic primitives and addressing scheme used throughout SHRINCS.

5.1 Hash Function

SHRINCS uses SHA-256 truncated to `n` bytes for all internal hash operations.

```
H(m) = SHA-256(m)[0:n]
```

5.2 Tweakable Hash Function

To prevent multi-target attacks and ensure domain separation between the various components, we employ the tweakable hash function construction defined in SPHINCS+ [**https://sphincs.org/data/sphincs+-paper.pdf**

```
Tw(PK.seed, ADRS, m) = H(PK.seed || ADRS || m)
```

This usage of tweaked hashing is critical. It ensures that a hash computed at one position in the Merkle tree cannot be substituted for a hash at another position, effectively mitigating generic tree-grafting attacks.

Note. For SHA-256, the compression function operates on 64-byte blocks. Since `PK.seed || ADRS` is $16 + 32 = 48$ bytes, we pad it to 64 bytes to optimize performance:

```
Tw(PK.seed, ADRS, m) = H(PK.seed || 0^16|| ADRS || m)
```

Having the first 64-byte precomputed, we reduce the hashing complexity to 1 block for each tweak function.

5.3 Pseudorandom Functions

SHRINCS uses two pseudorandom functions for different purposes: key derivation and getting the randomizer for the signature.

For deterministic derivation of secret key material (WOTS+C chain starting values, FORS secret keys):

```
PRF(PK.seed, ADRS, SK.seed) = Tw(PK.seed, ADRS, SK.seed)
```

The address structure ADRS ensures that each derived value is unique and bound to its position in the scheme.

For generating the randomizer `R` used in message hashing:

```
PRF_msg(SK.prf, PK.seed, opt_rand, m) = HMAC-SHA-256(SK.prf, PK.seed || opt_rand || m),
```

where `opt_rand` is an optional randomness (`n` bytes), can be `PK.seed` for deterministic signing.

5.4 Message Hash Function

SHRINCS uses different hashing strategies for WOTS+C and FORS+C, optimized for their respective use cases.

5.4.1 WOTS+C: Two-Phase Hashing

WOTS+C uses a two-phase approach for message hashing and grinding. This design is necessary because:

- User messages need randomized hashing (Phase 1 + Phase 2)
- Internal XMSS signatures sign tree roots that are already hashes (Phase 2 only)

Phase 1. Message Digest Computes a randomized hash of the user message:

```
H_msg(ADRS, R, PK.seed, PK.root, m) = SHA-256(ADRS || R || PK.seed || PK.root ||
                                        m)[0:n]
```

This is computed once per signature, producing an n-byte digest.

Phase 2. Grinding Hash Takes the digest and a counter:

```
H_grind(ADRS, digest, ctr) = SHA-256(ADRS || digest || ctr)[0:n]
```

This is computed repeatedly during grinding, but operates only on small fixed-size inputs (52 bytes), making iterations fast.

5.4.2 FORS+C: Single-Phase Hashing

FORS+C always signs user messages (never tree roots), so it uses a simpler single-phase approach with the counter included directly:

```
H_msg_fors(ADRS, R, ctr, PK.seed, PK.root, m) = SHA-256(ADRS || R || ctr || PK.seed ||
                                PK.root || m)[0:(k*a)/8]
```

5.5 Tweak Structure and Domain Separation

Tweaks are structured as follows to ensure uniqueness across all scheme components:

```
ADRS = layer || tree_address || type || words
```

where:

`layer`: (4 bytes) layer in hypertree (`0` = bottom, `d-1` = max)

`tree_address`: (12 bytes) tree address within layer

`type`: (4 bytes) address type

`words`: (12 bytes) type-dependent 12 bytes

A total `ADRS` size is 32 bytes.

5.5.1 Address Types

```
0x00: SF_WOTS_HASH: Stateful WOTS+C chain hashing
0x01: SF_WOTS_PK:   Stateful WOTS+C public key compression
0x02: SF_TREE:      Stateful tree internal nodes
0x03: SL_WOTS_HASH: Stateless WOTS+C chain hashing
0x04: SL_WOTS_PK:   Stateless WOTS+C public key compression
0x05: SL_TREE:      Stateless tree internal nodes
0x06: FORS_HASH:    FORS leaf hashing
0x07: FORS_TREE:    FORS tree internal nodes
0x08: FORS_PK:      FORS roots compression
0x09: WOTS_GRIND:   WOTS+C grinding
0x0A: FORS_GRIND:   FORS+C grinding
0x0B: H_MSG:        Message hash
0x0C: WOTS_PRF:     PRF for WOTS+C key derivation
0x0D: FORS_PRF:     PRF for FORS secret key derivation
0x0E: ROOT:         Root public key computation
```

Summarizing, `(0x00, 0x01, 0x02)` are stateful only types, `(0x03, 0x04, 0x05, 0x06, 0x07, 0x08, 0x0A)` are stateless only and `(0x09, 0x0B, 0x0C, 0x0D)` are shared types.

5.5.2 Address Manipulation Functions

```
new_ADRS():
     return toByte(0, 32)

setLayerAddress(ADRS, layer):
    ADRS[0:4] ← toByte(layer, 4)

setTreeAddress(ADRS, tree_addr):
    ADRS[4:16] ← toByte(tree_addr, 12)

setTypeAndClear(ADRS, type):
    ADRS[16:20] ← toByte(type, 4)
    ADRS[20:32] ← toByte(0, 12)

setKeyPairAddress(ADRS, keypair):
    ADRS[20:24] ← toByte(keypair, 4)

setChainAddress(ADRS, chain):
    ADRS[24:28] ← toByte(chain, 4)

setHashAddress(ADRS, hash):
    ADRS[28:32] ← toByte(hash, 4)

setTreeHeight(ADRS, height):
    ADRS[24:28] ← toByte(height, 4)

setTreeIndex(ADRS, index):
    ADRS[28:32] ← toByte(index, 4)
```

Note that `setTreeHeight` and `setChainAddress` both write to bytes `[24:28]`. These operations are context-dependent so there mustn't be a conflict.

5.5.3 Type-Dependent Words

6 WOTS+C

The standard Winternitz One-Time Signature (WOTS+) is the building block of most modern hash-based schemes. It works by encoding the message into a series of small integers and using them to iterate hash chains [**wots+**]. A significant portion of a WOTS+ signature size (approx 20%) is dedicated to "Checksum," which is necessary to prevent a forgery attack where an adversary extends the hash chains forward. WOTS+C (compressed WOTS+) eliminates the transmission of this checksum entirely, achieving a smaller signature size.

In WOTS+, the checksum C is calculated from the message digest D.

$$C = \sum_{i=1}^{l} (w - 1 - d_i)$$

This checksum is then signed using additional chains. In WOTS+C [**wots+c**], instead of calculating a checksum for a fixed message digest, the signer modifies the message digest until it possesses a specific, pre-determined checksum property. This is done by hashing the message with a random salt `R` (phase 1), then iterating over a counter `ctr` applied to the resulting digest (phase 2) until the output satisfies a global target condition.

Type	Name	word1 [20:24]	word2 [24:28]	word3 [28:32]
0x00	SF_WOTS_HASH	keypair	chain	hash
0x01	SF_WOTS_PK	keypair	0	0
0x02	SF_TREE	keypair	height	index
0x03	SL_WOTS_HASH	keypair	chain	hash
0x04	SL_WOTS_PK	keypair	0	0
0x05	SL_TREE	keypair	height	index
0x06	FORS_HASH	tree_idx	0	leaf_idx
0x07	FORS_TREE	tree_idx	height	index
0x08	FORS_PK	0	0	0
0x09	WOTS_GRIND	keypair	0	0
0x0A	FORS_GRIND	0	0	0
0x0B	H_MSG	0	0	0
0x0C	WOTS_PRF	keypair	chain	0
0x0D	FORS_PRF	tree_idx	0	leaf_idx
0x0E	ROOT	0	0	0

Table 1: Type-dependent interpretation of ADRS words (`bytes [20:32]`). For PRF (`0x0C`), field interpretation depends on context: WOTS uses (`keypair, chain, 0`), FORS uses (`tree_idx, 0, leaf_idx`). Compression addresses (`SF_WOTS_PK, SL_WOTS_PK, FORS_PK`) use only keypair/tree identification

6.1 The Target Condition

WOTS utilize a parameter `w` to define the number of hashchains and correspondingly their length. The message digest length is `n` bytes so `l = 8n/(log2(w))`. The values $d_{i\in[l]}$ are treated as integers in `0...(w-1)`.

Let `Swn` be a protocol parameter (often near the expected value, but may be chosen larger to trade signing/grinding cost for verification cost). If the hash output is uniform, the expected value of a single digit is `d_i = (w-1)/2`. For `l` digits, the expected `Swn` is:

$$S_{wn} = \sum_{i=0}^{l-1} d_i$$

By enforcing this condition during signing (including grinding), the verifier can simply assume the checksum is `Swn` without receiving it. If the reconstructed digest does not sum to `Swn`, the signature is invalid.

Additionally we can manipulate a target condition to decrease the cost of signature verification. The signature generation and verification operations together consist of `St = (w-1)*l` hashing operations. Increasing `S_wn` increases grinding cost but reduces the verification complexity to (`S_t - S_wn`) hashes. Concrete parameters are presented in Section 4.

6.2 Base-w Representation

WOTS represents a message digest in base `w` so that each digit is an integer in `[0...w-1]`. These digits directly determine the number of hashes to apply to each WOTS chain element during signing and verification. The `base_w()` function converts a byte string into an array of `out_len` base-w digits. In this specification we assume w is a power of two for efficiency, though this is not a strict requirement of the construction.

```
Algorithm: base_w(m, w, out_len)
    m: byte string
    w: Winternitz parameter
    out_len: output length
Output:
    b_w: array of out_len integers in [0, w-1]
```

```
1. in_idx ← 0
2. bits ← 0
3. total ← 0
4. b_w ← []
5. for i from 0 to (out_len - 1):
    a. if bits == 0:
        total ← m[in_idx]
        in_idx ← in_idx + 1
        bits ← 8
    b. bits ← bits - log2(w)
    c. b_w[i] ← (total >> bits) AND (w - 1)
6. return b_w
```

We provide optimized versions for `w = 256` and `w = 4` proposed by different SHRINCS instances. For `w=256`:

```
Algorithm: base_w256(m, out_len)
Input:
    m: byte string
    out_len: output length
Output:
    b_w: array of out_len integers in [0, 255]

1. b_w ← []
2. for i from 0 to (out_len - 1):
    a. b_w[i] ← m[i]
3. return b_w
```

For `w=4`:

```
Algorithm: base_w4(m, out_len)
Input:
    m: byte string
    out_len: output length
Output:
    b_w: array of out_len integers in [0,3]

1. b_w ← []
2. for j from 0 to (out_len/4 - 1):
    a. x ← m[j]
    b. b_w[4*j + 0] ← (x >> 6) AND 3
    c. b_w[4*j + 1] ← (x >> 4) AND 3
    d. b_w[4*j + 2] ← (x >> 2) AND 3
    e. b_w[4*j + 3] ← x AND 3
3. return b_w
```

6.3 Chain Function

WOTS derives signature components and public key material by iteratively applying the tweakable hash function along a Winternitz chain. The function `chain()` takes an n-byte input value `m` and applies `Tw(PK.seed, ADRS, ·)` exactly `steps` times, beginning at chain position `start`. For each iteration, the address field hash inside `ADRS` is set to the current chain position `i` (from `start` to `start + steps - 1`) to ensure domain separation between different chain steps. The output is the resulting n-byte value after advancing the chain by steps, which is used both for producing signature chain values (advancing from `0` to `msg[i]`) and for reconstructing end-of-chain values during verification (advancing from `msg[i]` to `w-1`).

The mode parameter determines which address types are used: SF for the stateful tree instance (SF_WOTS_HASH, SF_WOTS_PK, SF_TREE) or SL for the stateless instance (SL_WOTS_HASH, SL_WOTS_PK, SL_TREE).

```
Algorithm: chain(m, start, steps, PK.seed, ADRS)
Input:
    m: input
    start: starting index
    steps: number of iterations
    PK.seed: public seed
    ADRS: tweak (type already set to SF_WOTS_HASH or SL_WOTS_HASH)
Output:
    n-byte chain output

1. tmp ← m
2. for i from start to (start + steps - 1):
    a. setHashAddress(ADRS, i)
    b. tmp ← Tw(PK.seed, ADRS, tmp)
3. return tmp
```

6.4 Key Generation

To create a WOTS public key for a given keypair index, we derive one secret starting value per chain using PRF, then advance each value to the end of its chain. The resulting array of end-of-chain values is compressed into a single n-byte WOTS public key via the tweakable hash. The compression uses address type SF_WOTS_PK (stateful) or SL_WOTS_PK (stateless).

```
Algorithm: wots_PKgen(SK.seed, PK.seed, ADRS, keypair, mode)
Input:
    SK.seed: secret seed
    PK.seed: public seed
    ADRS: tweak
    keypair: keypair index
    mode: SF or SL
Output:
    pk: compressed WOTS+C public key

    if mode == SF:
        WOTS_HASH ← SF_WOTS_HASH
        WOTS_PK   ← SF_WOTS_PK
        TREE      ← SF_TREE
    else:
        WOTS_HASH ← SL_WOTS_HASH
        WOTS_PK   ← SL_WOTS_PK
        TREE      ← SL_TREE

1. tmp ← []
2. for i from 0 to l - 1:
    a. setTypeAndClear(ADRS, WOTS_PRF)
    b. setKeyPairAddress(ADRS, keypair)
    c. setChainAddress(ADRS, i)
    d. sk_i ← PRF(PK.seed, ADRS, SK.seed)
    e. setTypeAndClear(ADRS, WOTS_HASH)
    f. setKeyPairAddress(ADRS, keypair)
    g. setChainAddress(ADRS, i)
    h. tmp[i] ← chain(sk_i, 0, w - 1, PK.seed, ADRS)
3. setTypeAndClear(ADRS, WOTS_PK)
```

```
4. setKeyPairAddress(ADRS, keypair)
5. pk ← Tw(PK.seed, ADRS, tmp[0] || ... || tmp[l-1])
6. return pk
```

6.5 Messsage Digest Grinding

WOTS+C uses a two-phase hashing approach (see Section 5.4.1):

1. Phase 1 (`H_msg`): Computes a randomized digest of the user message. This is performed once per signature and produces an n-byte digest. For internal XMSS signatures (which sign tree roots that are already hashes), Phase 1 is skipped and the tree root is used directly as the digest.

2. Phase 2 (`H_grind`): Takes the Phase 1 digest (or tree root) and a counter, producing a candidate n-byte output. This is iterated during grinding until the target condition is met.

The function `wots_grind()` takes a pre-computed digest and increments `ctr` until the base-w digits of the Phase 2 output satisfy the target sum `S_wn`. It returns the found `ctr` and the corresponding digit array.

```
Algorithm: wots_grind(digest, ADRS)
Input:
    digest: a digest from Phase 1 H_msg, or a tree root for internal signatures
    ADRS: tweak
Output:
    msg: array of l integers in [0, w-1]
    ctr: final counter value used

1. setTypeAndClear(ADRS, WOTS_GRIND)
2. for ctr from 0 to 2^32 - 1:
    a. grind_out ← H_grind(ADRS, digest, toByte(ctr, c))
    b. msg ← base_w(grind_out, w, l)
    c. sum ← 0
    d. for i from 0 to (l - 1):
        sum ← sum + msg[i]
    e. if sum == S_wn:
        return (msg, ctr)
3. abort with error  // caller should resample R and retry
```

Oleks: to add expected grinding complexity for different approaches

Additionally we need to define a func which allows to calculate the digest without grinding (by providing the `ctr` value directly).

```
Algorithm: wots_digest(digest, ctr, ADRS)
Input:
    digest: digest from Phase 1 H_msg, or a tree root
    ctr: counter value
ADRS: tweak

Output:
    msg: array of l integers in [0, w-1]
    valid: boolean indicating if sum == S_wn

1. setTypeAndClear(ADRS, WOTS_GRIND)
2. grind_out ← H_grind(ADRS, digest, toByte(ctr, c))
3. msg ← base_w(grind_out, w, l)
4. sum ← 0
5. for i from 0 to l - 1:
    sum ← sum + msg[i]
```

```
6. valid ← (sum == S_wn)
7. return (msg, valid)
```

6.6 Signature Generation

To sign a user message, WOTS+C first computes the phase 1 digest, then grinds to find a valid counter. For each chain, it reveals the chain value after exactly `msg[i]` steps (starting from the secret-derived chain start). The signature carries `R`, the grinding counter `ctr`, and the `l` chain values.

For internal XMSS signatures (signing tree roots at layers `1..d-1` of the hypertree), phase 1 is skipped: the tree root serves directly as the digest input to `wots_grind`. In this case `R` is still generated and included in the signature to maintain a uniform format, but it does not affect the digest computation.

```
Algorithm: wots_sign(m, SK.seed, SK.prf, PK.seed, PK.root, ADRS, keypair, mode, is_internal)
Input:
    m: message
    SK.seed: secret seed
    SK.prf: PRF key for randomness
    PK.seed: public seed
    PK.root: public key root
    ADRS: tweak
    keypair: keypair index
    mode: SF or SL
    is_internal: boolean (true for internal XMSS tree-root signatures)

Output:
    sig: WOTS+C signature (R || ctr || chain_values)

    if mode == SF:
        WOTS_HASH ← SF_WOTS_HASH
    else:
        WOTS_HASH ← SL_WOTS_HASH

1. opt_rand ← random(n) or PK.seed     // randomized or deterministic
2. R ← PRF_msg(SK.prf, PK.seed, opt_rand, m)
3. if is_internal:
        digest ← m
    else:
        setTypeAndClear(ADRS, H_MSG)
        digest ← H_msg(ADRS, R, PK.seed, PK.root, m)
4. (msg, ctr) ← wots_grind(digest, ADRS)
5. sig ← R || toByte(ctr, c)
6. for i from 0 to (l - 1):
    a. setTypeAndClear(ADRS, WOTS_PRF)
    b. setKeyPairAddress(ADRS, keypair)
    c. setChainAddress(ADRS, i)
    d. sk_i ← PRF(PK.seed, ADRS, SK.seed)
    e. setTypeAndClear(ADRS, WOTS_HASH)
    f. setKeyPairAddress(ADRS, keypair)
    g. setChainAddress(ADRS, i)
    h. sig ← sig || chain(sk_i, 0, msg[i], PK.seed, ADRS)
7. return sig
```

6.7 Signature Verification (Public Key Recovery)

Verification reconstructs the WOTS+C public key implied by the signature. It first recomputes the message digest using the two-phase approach (or directly from the tree root for internal signatures),

then verifies the target sum. For each chain element, the verifier continues hashing forward from the signature's provided intermediate value for `(w - 1 - msg[i])` steps to reach the end-of-chain value. Compressing all end-of-chain values yields the recovered WOTS+C public key.

```
Algorithm: wots_pkFromSig(sig, m, PK.seed, PK.root, ADRS, keypair, mode, is_internal)
Input:
    sig: WOTS+C signature (R || ctr || chain_values)
    m: signed message (user message or tree root)
    PK.seed: public seed
    PK.root: root public key
    ADRS: tweak
    keypair: keypair index
    mode: SF or SL
    is_internal: boolean (true for internal XMSS tree-root signatures)
Output:
    pk: recovered public key (n bytes), or $\bot$ if invalid

    if mode == SF:
        WOTS_HASH ← SF_WOTS_HASH
        WOTS_PK   ← SF_WOTS_PK
    else:
        WOTS_HASH ← SL_WOTS_HASH
        WOTS_PK   ← SL_WOTS_PK

1. R ← sig[0 : R_SIZE]
2. ctr ← sig[R_SIZE : R_SIZE + c]
3. chains_start ← R_SIZE + c
4. if is_internal:
        digest ← m
    else:
        setTypeAndClear(ADRS, H_MSG)
        digest ← H_msg(ADRS, R, PK.seed, PK.root, m)
5. (msg, valid) ← wots_digest(digest, ctr, ADRS)
6. if not valid:
    return $\bot$
7. tmp ← []
8. for i from 0 to (l - 1):
    a. setTypeAndClear(ADRS, WOTS_HASH)
    b. setKeyPairAddress(ADRS, keypair)
    c. setChainAddress(ADRS, i)
    d. sig_i ← sig[chains_start + i * n : chains_start + (i+1) * n]
    e. tmp[i] ← chain(sig_i, msg[i], w - 1 - msg[i], PK.seed, ADRS)
9. setTypeAndClear(ADRS, WOTS_PK)
10. setKeyPairAddress(ADRS, keypair)
11. pk ← Tw(PK.seed, ADRS, tmp[0] || ... || tmp[l-1])
12. return pk
```

6.8 Contribution to the Signature Size and Verification Complexity

Each WOTS+C signature contributes the following to the enclosing signature:

- `R`: `R_SIZE` bytes
- `ctr`: `c` bytes
- chain values: `l*n` bytes

Total: `R_SIZE + c + l*n` bytes (292 bytes for SHRINCS-B and 1060 bytes for SHRINCS-L)

The verification cost is `(w-1)*l - S_wn + 1` hashes per WOTS+C instance (2041 hashes for SHRINCS-B and 53 hashes for SHRINCS-L): the verifier completes each chain forward by `(w - 1 - msg[i])` steps, and the sum of all forward steps is `(w-1)*l - S_wn` (since the sum of `msg[i]` values equals `S_wn`). One additional hash is needed for the public key compression.

7 XMSS

SHRINCS employs two distinct Merkle tree constructions: a standard balanced XMSS tree for the stateless hypertree, and an Unbalanced XMSS (UXMSS) tree optimized for the stateful component (specified in Section 8).

The stateless signature mode uses a SPHINCS+-style hypertree consisting of `d` layers, where each layer contains standard balanced XMSS trees. Each XMSS tree authenticates `2^h'` WOTS+C key pairs, where `h' = h_sl / d` is the height of trees within each layer.

7.1 XMSS Tree Structure

An XMSS tree of height `h'` is a complete binary Merkle tree where:

- Leaves: The `2^h'` leaves are WOTS+C public keys, computed via `wots_PKgen` with mode `SL`.
- Internal nodes: Each internal node is the tweakable hash of its two children, using address type `SL_TREE`.
- Root: The tree root serves as either the layer's contribution to the hypertree or (for the top layer) as `PK.sl`.

Layer indexing within the stateless hypertree is the following:

- Layer 0: Bottom layer (is used to sign the FORS+C public keys)
- Layer d-1: Top layer (root is `PK.sl`)

7.2 XMSS TreeHash

The `xmss_treehash` function computes subtree roots within an XMSS tree. It takes a target height and a leftmost leaf index, then recursively builds the subtree by computing leaves (WOTS+C public keys) at height 0 and hashing pairs of children to form internal nodes. This function is the core building block used both for computing the full tree root (by calling it with `target_height = h'` and `start_idx = 0`) and for generating individual sibling nodes needed in authentication paths (with arbitrary `start_idx`).

```
Algorithm: xmss_treehash(SK.seed, PK.seed, ADRS, target_height, start_idx)
Input:
    SK.seed: secret seed
    PK.seed: public seed
    ADRS: tweak
    target_height: height of output node (0 = leaf level)
    start_idx: leftmost leaf index in the subtree
Output:
    node: hash value

1. if target_height == 0:
    a. pk ← wots_PKgen(SK.seed, PK.seed, ADRS, start_idx, SL)
    b. return pk
2. left ← xmss_treehash(SK.seed, PK.seed, ADRS, target_height - 1, start_idx)
3. right ← xmss_treehash(SK.seed, PK.seed, ADRS, target_height - 1, start_idx +
                                                    + 2^(target_height - 1))
4. setTypeAndClear(ADRS, SL_TREE)
5. setTreeHeight(ADRS, target_height)
6. setTreeIndex(ADRS, start_idx >> target_height)
7. return Tw(PK.seed, ADRS, left || right)
```

Note: `start_idx` need not be zero. When computing a full tree root, it is called with `start_idx = 0`. When computing a sibling node for an authentication path, `start_idx` is the leftmost leaf index of that sibling's subtree.

7.3 XMSS Root Computation

The `xmss_root` function computes the root of an entire XMSS tree by invoking `xmss_treehash` with the full tree height and starting from leaf index 0. The resulting root hash represents the public key for this XMSS tree instance, which either becomes `PK.sl` (for the top hypertree layer) or is signed by the layer above (for lower layers).

```
Algorithm: xmss_root(SK.seed, PK.seed, ADRS, h_prime)
Input:
    SK.seed: secret seed
    PK.seed: public seed
    ADRS: tweak
    h_prime: tree height
Output:
    root: tree root (n bytes)

1. return xmss_treehash(SK.seed, PK.seed, ADRS, h_prime, 0)
```

7.4 XMSS Authentication Path Generation

For a leaf at index `idx`, the authentication path consists of the sibling nodes at each level from the leaf up to (but not including) the root. The authentication path contains `h'` nodes, each of size `n` bytes, allowing a verifier to reconstruct the path from the leaf to the root.

At height `h`, the sibling subtree's leftmost leaf index is determined by flipping the h-th bit of `idx` (using XOR) and clearing all lower bits. The sibling node is then computed by calling `xmss_treehash` on the subtree rooted at that position.

```
Algorithm: xmss_auth_path(SK.seed, PK.seed, ADRS, h_prime, idx)
Input:
    SK.seed, PK.seed: seeds
    ADRS: tweak
    h_prime: tree height
    idx: leaf index being authenticated
Output:
auth: authentication path (h_prime * n bytes)

1. auth ← []
2. for h from 0 to h_prime - 1:
    a. sibling_start ← (idx XOR (1 << h)) AND (0xFFFFFFFF << h)
    b. auth ← auth || xmss_treehash(SK.seed, PK.seed, ADRS, h, sibling_start)
3. return auth
```

Note: The expression `(idx XOR (1 << h)) AND (0xFFFFFFFF << h)` flips the `h`-th bit of `idx` to identify the sibling, then clears the lower `h` bits to obtain the leftmost leaf index of the sibling's subtree of height `h`.

7.5 XMSS Public Key Recovery from Signature

Given a WOTS+C signature and authentication path, this algorithm recovers the XMSS tree root. First, it recovers the WOTS+C public key (the leaf value) from the WOTS+C signature. Then, starting from the leaf, it iteratively combines the current node with the corresponding authentication path node to compute the parent, continuing until reaching the root. At each height `h`, the `h`-th bit of idx determines

whether the current node is the left child (`bit = 0`) or right child (`bit = 1`), which dictates the ordering when hashing the pair.

The message `m` being verified is either a FORS+C public key (at layer 0) or a lower-layer tree root (at layers `1...d-1`). In both cases, m is already a hash, so `wots_pkFromSig` is called with `is_internal = true` (Phase 1 of message hashing is skipped; see Section 6.7).

```
Algorithm: xmss_pkFromSig(wots_sig, auth, m, PK.seed, PK.root, ADRS, h_prime, idx)
Input:
    wots_sig: WOTS+C signature
    auth: authentication path (h_prime * n bytes)
    m: signed message (FORS+C public key or lower-layer tree root)
    PK.seed: public seed
    PK.root: root public key
    ADRS: tweak
    h_prime: tree height
    idx: leaf index
Output:
    root: computed tree root (n bytes), or $\bot$ if invalid

1. pk ← wots_pkFromSig(wots_sig, m, PK.seed, PK.root, ADRS, idx, SL, true)
2. if pk == $\bot$:
    return $\bot$
3. node ← pk
4. for h from 0 to h_prime - 1:
    a. setTypeAndClear(ADRS, SL_TREE)
    b. setTreeHeight(ADRS, h + 1)
    c. setTreeIndex(ADRS, idx >> 1)
    d. auth_node ← auth[h * n : (h + 1) * n]
    e. if (idx AND 1) == 0:
            node ← Tw(PK.seed, ADRS, node || auth_node)
        else:
            node ← Tw(PK.seed, ADRS, auth_node || node)
f. idx ← idx >> 1
5. return node
```

7.6 XMSS Signature Generation (Stateless Layer)

To sign a message within the stateless hypertree, the XMSS signature generation produces a WOTS+C signature on the message followed by the authentication path for the corresponding leaf. The message being signed is typically either a FORS+C public key (at layer 0) or the root of a lower-layer XMSS tree (at layers 1 through `d-1`). In both cases the message is already a hash value, so `wots_sign` is called with `is_internal = true`.

The leaf index idx determines which WOTS+C key pair is used; this index is determined from the message digest during stateless signing (see Section 10.5).

```
Algorithm: xmss_sign(m, SK.seed, SK.prf, PK.seed, PK.root, ADRS, h_prime, idx)
Input:
    m: message to sign (lower-layer root or FORS public key)
    SK.seed: secret seed
    SK.prf: PRF key
    PK.seed: public seed
    PK.root: root public key
    ADRS: tweak
    h_prime: tree height
    idx: leaf index to use
```

```
Output:
    sig: XMSS signature (WOTS+C signature || authentication path)

1. wots_sig ← wots_sign(m, SK.seed, SK.prf, PK.seed, PK.root, ADRS, idx, SL, true)
2. auth ← xmss_auth_path(SK.seed, PK.seed, ADRS, h_prime, idx)
3. return wots_sig || auth
```

7.7 Contribution to Signature Size and Verification Complexity

Each XMSS signature consists of a WOTS+C signature plus an authentication path of `h'` nodes: `R_SIZE + c + (l + h') * n` bytes (420 bytes for SHRINCS-B and 1188 bytes for SHRINCS-L).

The full stateless hypertree consists of d = 5 layers, giving:

- 2100 bytes and 10245 hashes for SHRINCS-B
- 5940 bytes and 305 hashes for SHRINCS-L

Note: This excludes the FORS+C contribution (see Section 9). The total stateless signature size and verification cost are the sum of FORS+C and hypertree components, as presented in Section 3.2.2.

8 UXMSS. The Stateful Tree

Standard XMSS uses a balanced binary tree, which minimizes the maximum path length but imposes a logarithmic cost on every signature. For `N` signatures, every signature has size proportional to `log_2(N)`. SHRINCS utilizes an Unbalanced Merkle Tree (specifically a "Right-Skewed" tree) to optimize for the typical Bitcoin use case: a wallet that performs very few transactions in its lifecycle.

In this topology the `q`-th signature requires a path of length `q`. This linear growth is superior to logarithmic growth for small `q`. For `q=1`, the path is just 16 bytes. Even for `q=10`, the path is $10 \times 16 = 160$ bytes, which is extremely small compared to the fixed $\approx$ 2.5KB of a balanced tree or the $\approx$ 8KB of SLH-DSA.

8.1 UXMSS Tree Structure

The UXMSS tree is constructed recursively as a right-skewed binary tree with `h_sf + 1` leaves. To describe the tree structure, we use "level" to denote depth from the root, with level 0 at the root and level `h_sf - 1` at the bottom internal node. Note that this UXMSS convention is the inverse of the convention used in the balanced XMSS trees of the stateless component (Section 7). However, the ADRS height field for internal nodes is set to (`h_sf - level`), so both constructions use increasing height values toward the root. The UXMSS tree is constructed as follows:

- Level 0 is the root public key `PK.sf`.
- Level `i` (for `0 <= i <= h_sf-2`, internal):
  - Left child: WOTS+C public key for signature `q=i+1`
  - Right child: Root of subtree for signatures `q>i+1`
- Level `hsf-1` (bottom)
  - Left child: WOTS+C public key for signature `q=hsf`
  - Right child: WOTS+C public key for signature `q=hsf+1`

This means the tree has `h_sf + 1` leaves (one per valid signature index `q` $\in$ {`1, ..., h_sf + 1`}), and every internal node has a WOTS+C leaf as its left child and a UXMSS subtree (or a second leaf at the bottom) as its right child.

8.2 UXMSS TreeHash

The `uxmss_treehash` function computes subtree roots within the right-skewed UXMSS tree. Unlike balanced XMSS where treehash recursively computes both subtrees symmetrically, UXMSS treehash follows the right-skewed structure: the left child is always a WOTS+C public key (a leaf), while the right child is either another UXMSS subtree (for internal levels) or a second WOTS+C public key (at the bottom level). The level parameter indicates depth from the root, with level 0 being the root and level `h_sf - 1` being the bottom internal node.

All UXMSS operations use the stateful address types (`SF_WOTS_HASH`, `SF_WOTS_PK`, `SF_TREE`) and are fixed at `layer = 0`, `tree_address = 0` (since there is exactly one stateful tree per SHRINCS key pair).

```
Algorithm: uxmss_treehash(SK.seed, PK.seed, ADRS, level)
Input:
    SK.seed: secret seed
    PK.seed: public seed
    ADRS: tweak (layer = 0, tree_address = 0)
    level: depth from root (0 = root, h_sf - 1 = bottom internal node)
Output:
    node: hash value (n bytes)

1. left ← wots_PKgen(SK.seed, PK.seed, ADRS, level + 1, SF)
2. if level == h_sf - 1:
        right ← wots_PKgen(SK.seed, PK.seed, ADRS, h_sf + 1, SF)
    else:
        right ← uxmss_treehash(SK.seed, PK.seed, ADRS, level + 1)
3. setTypeAndClear(ADRS, SF_TREE)
4. setTreeHeight(ADRS, h_sf - level)
5. setTreeIndex(ADRS, 0)
6. return Tw(PK.seed, ADRS, left || right)
```

Note: The keypair index passed to `wots_PKgen` corresponds to the signature number `q`. At level `i`, the left child is the leaf for `q = i + 1`. At the bottom level (`h_sf - 1`), the right child is the leaf for `q = h_sf + 1`.

8.3 UXMSS Root Computation

The `uxmss_root` function computes `PK.sf`, the root of the stateful UXMSS tree. It initializes the address structure with layer 0 and tree address 0 (since there is only one stateful tree), then invokes `uxmss_treehash` starting from level 0. The resulting root, combined with `PK.sl` from the stateless tree, forms the basis for `PK.root`.

```
Algorithm: uxmss_root(SK.seed, PK.seed, ADRS)
Input:
    SK.seed: secret seed
    PK.seed: public seed
    ADRS: tweak
Output:
    PK.sf: stateful public key

1. setLayerAddress(ADRS, 0)
2. setTreeAddress(ADRS, 0)
3. return uxmss_treehash(SK.seed, PK.seed, ADRS, 0)
```

8.4 UXMSS Authentication Path Generation

For signature number `1 <= q <= hsf + 1`, the authentication path contains exactly `q` sibling nodes needed to reconstruct `PK.sf` from the WOTS+C public key at position `q`. The structure of the path

differs based on whether `q <= hsf` or `q = hsf + 1`. For `q <= hsf`, the first sibling is either a subtree root (if `q < hsf`) or the rightmost WOTS+C key (if `q = hsf`), followed by q-1 WOTS+C public keys ascending toward the root. For `q = hsf + 1` (the rightmost leaf), all hsf siblings are WOTS+C public keys. This asymmetric structure reflects the right-skewed nature of UXMSS.

Mike: I think we still can use idx variable, and just use it as a pointer to the leaf, counted from left to right.

Oleks: we can, but I think the semantics of `q` and `idx` are different. `idx` is a position in a symmetric tree where all leaves are equivalent, the auth path is always `h'` nodes regardless of which `idx` we pick. `q` is the signature counter that directly determines the auth path length and structure

```
Algorithm: uxmss_auth_path(SK.seed, PK.seed, ADRS, q)
Input:
    SK.seed, PK.seed: seeds
    ADRS: address
    q: signature number (1 to hsf + 1)
Output:
    auth: authentication path (q * n bytes)

1. auth ← []
2. setLayerAddress(ADRS, 0)
3. setTreeAddress(ADRS, 0)
4. if q <= hsf:
        if q == hsf:
            auth ← auth || wots_PKgen(SK.seed, PK.seed, ADRS, hsf + 1, SF)
        else:
            auth ← auth || uxmss_treehash(SK.seed, PK.seed, ADRS, q)
        for i from 1 to q - 1:
            auth ← auth || wots_PKgen(SK.seed, PK.seed, ADRS, q - i, SF)
5. else:
        for i from 0 to hsf - 1:
            auth ← auth || wots_PKgen(SK.seed, PK.seed, ADRS, hsf - i, SF)
6. return auth
```

Note: For `q <= h_sf`, the authentication path has exactly q nodes. For `q = h_sf + 1`, it has `h_sf` nodes, since the rightmost leaf shares its parent's other child (a WOTS+C key) rather than a subtree root.

8.5 UXMSS Public Key Recovery from Signature

Given a WOTS+C signature and authentication path, this algorithm reconstructs `PK.sf`. First, it recovers the WOTS+C public key from the signature. Then, starting from this leaf, it iteratively combines the current node with authentication path nodes to compute parent nodes until reaching the root.

The combination order follows from the tree structure:

- Case `q <= h_sf`: The recovered key is a left child. The first combination pairs the recovered key (left) with the first auth node (right). All subsequent combinations have the auth node (a WOTS+C key from a lower `q` value) on the left and the current node on the right.
- Case `q = h_sf + 1`: The recovered key is the rightmost leaf (a right child). All auth nodes are WOTS+C keys on the left.

```
Algorithm: uxmss_pkFromSig(wots_sig, auth, m, PK.seed, PK.root, ADRS, q)
Input:
    wots_sig: WOTS+C signature
    auth: authentication path
```

```
    m: signed message
    PK.seed: public seed
    PK.root: root public key
    ADRS: address
    q: signature number
Output:
    root: computed PK.sf, or $\bot$ if invalid

1. setLayerAddress(ADRS, 0)
2. setTreeAddress(ADRS, 0)
3. pk ← wots_pkFromSig(wots_sig, m, PK.seed, PK.root, ADRS, q, SF, false)
4. if pk == $\bot$:
    return $\bot$
5. node ← pk
6. setTypeAndClear(ADRS, SF_TREE)
7. if q <= h_sf:
    a. setTreeHeight(ADRS, h_sf - (q - 1))
    b. setTreeIndex(ADRS, 0)
    c. node ← Tw(PK.seed, ADRS, node || auth[0 : n])
    for i from 1 to q - 1:
        a. setTreeHeight(ADRS, h_sf - (q - 1 - i))
        b. setTreeIndex(ADRS, 0)
        c. node ← Tw(PK.seed, ADRS, auth[i * n : (i + 1) * n] || node)
8. else:
    for i from 0 to h_sf - 1:
        a. setTreeHeight(ADRS, i + 1)
        b. setTreeIndex(ADRS, 0)
        c. node ← Tw(PK.seed, ADRS, auth[i * n : (i + 1) * n] || node)
9. return node
```

Note: In step 3, `is_internal = false` because the stateful UXMSS signs user messages directly (requiring phase 1 + phase 2 hashing). This differs from the stateless XMSS layers which always sign tree roots or FORS public keys (`is_internal = true`).

8.6 UXMSS Signature Generation

To sign a message using the stateful path, UXMSS signature generation produces a WOTS+C signature followed by the authentication path for position `q`. The signature number `q` must be tracked as state and incremented after each use to ensure one-time signature security. As noted in Section 1.2, the state counter should be persisted before generating the signature to prevent key reuse in case of interruption.

The resulting signature size is `|wots_sig| + q*n` bytes, which grows linearly with `q`. For the first signature (q = 1), this yields the minimal signature that makes SHRINCS attractive for Bitcoin applications.

```
Algorithm: uxmss_sign(m, SK.seed, SK.prf, PK.seed, PK.root, ADRS, q)
Input:
    m: message
    SK.seed: secret seed
    SK.prf: PRF key
    PK.seed: public seed
    PK.root: root public key
    ADRS: address
    q: signature number
Output:
    sig: UXMSS signature (WOTS+C signature || authentication path)
```

```
1. setLayerAddress(ADRS, 0)
2. setTreeAddress(ADRS, 0)
3. wots_sig ← wots_sign(m, SK.seed, SK.prf, PK.seed, PK.root, ADRS, q, SF, false)
4. auth ← uxmss_auth_path(SK.seed, PK.seed, ADRS, q)
5. return wots_sig || auth
```

8.7 Contribution to Signature Size and Verification Complexity

Each UXMSS signature consists of a WOTS+C signature plus an authentication path whose length depends on the signature index `q`:

```
total = R_SIZE + c + l*n + q*n
```

For concrete instances and signatures' number:

```
SHRINCS-B:
    q=1:        32 + 4 + 16*16 + 16 =       308 bytes
    q=10:       32 + 4 + 16*16 + 16*10 =    452 bytes
    q_max=226:  32 + 4 + 16*16 + 16*226 =   3908 bytes // < 3992 bytes (stateless)
SHRINCS-L:
    q=1:        32 + 4 + 64*16 + 16 =       1076 bytes
    q=10:       32 + 4 + 64*16 + 16*10 =    1220 bytes
    q_max=418:  32 + 4 + 64*16 + 16*418 =   7748 bytes // < 7832 bytes (stateless)
```

Verification cost per UXMSS signature is calculated as WOTS+C verification cost + `q` hashes.

9 FORS+C

FORS (Forest of Random Subsets) is a few-time signature scheme used at the bottom of the stateless hypertree. FORS allows to sign a limited number of messages, when the security degrades with each use. Standard FORS signs a message by revealing `k` secret values, one from each of `k` binary trees of height `a`. The message digest determines which leaf to reveal from each tree. FORS+C applies the similar grinding optimization as WOTS+C: by searching for a digest where the last tree's index equals 0, the authentication path and leaf for that tree can be omitted entirely from the signature.

9.1 FORS Structure

FORS consists of `k` binary trees, each of height `a`:

- Each tree has 2^a leaves
- Each leaf is derived from `SK.seed` using PRF with address type `FORS_PRF`
- The message digest is split into `k` chunks of `a` bits each
- Each chunk selects one leaf index from the corresponding tree

The FORS public key is the tweakable hash of the tree roots. With the +C grinding optimization, the last tree's root is excluded: `FORS+C PK = Tw(PK.seed, ADRS, root_0 || ... || root_k-2)`, using address type `FORS_PK`. This is valid because grinding guarantees `indices[k-1] == 0`, so the verifier can independently recompute `root_k-1` if needed (though it is not used in the public key computation).

9.2 Grinding Condition

FORS+C uses the single-phase message hash `H_msg_fors` (Section 5.4.2):

```
digest = H_msg(ADRS, R, ctr, PK.seed, PK.root, m)
condition: digest[last a bits] == 0
```

Expected grinding cost is $\approx 2^a$ iterations. With a = 22, this is approximately 4 million iterations. Since `H_msg_fors` operates on fixed-size inputs, each iteration is a single SHA-256 call.

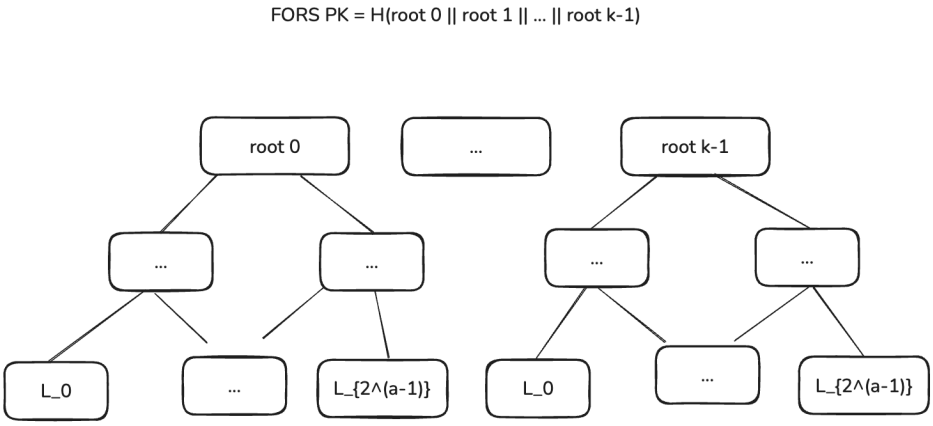

Figure 2: FORS architecture

9.3 Message to Indices

The `fors_msg_to_indices` function converts a message digest into `k` leaf indices, one for each FORS tree. Each index is extracted as a consecutive a-bit chunk from the digest. The indices are integers in the range `[0, 2^a - 1]`.

```
Algorithm: fors_msg_to_indices(digest)
Input:
    digest: message digest
Output:
    indices: array of k integers in [0, 2^a - 1]

1. indices ← []
2. for i from 0 to k - 1:
    a. indices[i] ← extract_bits(digest, i * a, a)
3. return indices
```

Note: The message digest also encodes which FORS instance is used, as determined by the tree and leaf indices parsed from the stateless signature digest (Section 10.5). The output size of `H_msg_fors` must be sufficient to encode all `k` indices: at least `roundup((k*a) / 8)` bytes.

9.4 Grinding

The `fors_grind` function searches for a counter value where the last tree's index equals 0. This eliminates the need to include the authentication path and leaf for tree `k-1` in the signature.

```
Algorithm: fors_grind(m, R, PK.seed, PK.root, ADRS)
Input:
    m: message
    R: randomness
    PK.seed: public seed
    PK.root: public key root
    ADRS: tweak
Output:
    indices: array of k integers
    ctr: counter value

1. setTypeAndClear(ADRS, FORS_GRIND)
2. for ctr from 0 to 2^32 - 1:
```

```
    a. digest ← H_msg_fors(ADRS, R, ctr, PK.seed, PK.root, m)
    b. indices ← fors_msg_to_indices(digest)
    c. if indices[k-1] == 0:
        return (indices, ctr)
3. abort with error
```

9.5 Secret Key Generation

The `fors_SKgen` function derives a secret leaf value for a specific tree and leaf index. Each leaf's secret is deterministically generated using PRF with the tree index and leaf index encoded in the address structure via `FORS_PRF`. This ensures all 2^a leaves across all `k` trees can be regenerated from `SK.seed` without storing them.

```
Algorithm: fors_SKgen(SK.seed, PK.seed, ADRS, tree_idx, leaf_idx)
Input:
    SK.seed: secret seed
    PK.seed: public seed
    ADRS: tweak
    tree_idx: tree index (0 to k-1)
    leaf_idx: leaf index within tree
Output:
    sk: secret value (n bytes)

1. setTypeAndClear(ADRS, PRF)
2. setKeyPairAddress(ADRS, tree_idx)
3. setTreeIndex(ADRS, leaf_idx)
4. sk ← PRF(PK.seed, ADRS, SK.seed)
5. return sk
```

9.6 TreeHash

The `fors_treehash` function recursively computes FORS subtree roots. At height 0 (leaf level), it generates the secret key and hashes it to produce the leaf node using address type `FORS_HASH`. At higher levels, it recursively computes left and right children, then hashes their concatenation using `FORS_TREE`. This function is used both for computing full tree roots and for generating individual sibling nodes needed in authentication paths.

Note: The structure is analogous to `xmss_treehash` (Section 7.2), with the difference being the leaf computation (FORS uses hashed secret keys rather than WOTS+C public keys).

```
Algorithm: fors_treehash(SK.seed, PK.seed, ADRS, tree_idx, target_height, start_idx)
Input:
    SK.seed, PK.seed: seeds
    ADRS: tweak
    tree_idx: tree index (0 to k-1)
    target_height: height of output node (0 = leaf level)
    start_idx: leftmost leaf index in subtree
Output:
    node: hash value

1. if target_height == 0:
    a. sk ← fors_SKgen(SK.seed, PK.seed, ADRS, tree_idx, start_idx)
    b. setTypeAndClear(ADRS, FORS_HASH)
    c. setKeyPairAddress(ADRS, tree_idx)
    d. setTreeHeight(ADRS, 0)
    e. setTreeIndex(ADRS, start_idx)
    f. return Tw(PK.seed, ADRS, sk)
```

```
2. left ← fors_treehash(SK.seed, PK.seed, ADRS, tree_idx, target_height - 1, start_idx)
3. right ← fors_treehash(SK.seed, PK.seed, ADRS, tree_idx, target_height - 1, start_idx + 2^(tar
4. setTypeAndClear(ADRS, FORS_TREE)
5. setKeyPairAddress(ADRS, tree_idx)
6. setTreeHeight(ADRS, target_height)
7. setTreeIndex(ADRS, start_idx >> target_height)
8. return Tw(PK.seed, ADRS, left || right)
```

9.7 FORS Authentication path

The `fors_auth_path` function generates the a sibling nodes needed to authenticate a revealed leaf up to its tree root. At each height h, the sibling is the node that, when combined with the current path node, produces the parent. The sibling index is computed by XORing the leaf index with `2^h`, which flips the h-th bit to identify the sibling's subtree.

```
Algorithm: fors_auth_path(SK.seed, PK.seed, ADRS, tree_idx, leaf_idx)
Input:
    SK.seed, PK.seed: seeds
    ADRS: tweak
    tree_idx: tree index (0 to k-1)
    leaf_idx: leaf index being revealed
Output:
    auth: authentication path (a * n bytes)

1. auth ← []
2. for h from 0 to a - 1:
    a. sibling_start ← (leaf_idx XOR 2^h) AND (0xFFFFFFFF << h)
    b. auth ← auth || fors_treehash(SK.seed, PK.seed, ADRS, tree_idx, h, sibling_start)
3. return auth
```

9.8 Signature generation

The `fors_sign` function produces a FORS+C signature. Due to grinding, the last tree's index is guaranteed to be 0, so tree `k-1` is entirely excluded from the signature. The signature reveals `k-1` leaves with their authentication paths. Signature format: `R || ctr || (sk_0 || auth_0) || ... || (sk_k-2 || auth_k-2)`

```
Algorithm: fors_sign(m, SK.seed, SK.prf, PK.seed, PK.root, ADRS)
Input:
    m: message
    SK.seed: secret seed
    SK.prf: PRF key
    PK.seed: public seed
    PK.root: public key root
    ADRS: tweak
Output:
    sig: FORS+C signature

1. opt_rand ← random(n) or PK.seed
2. R ← PRF_msg(SK.prf, PK.seed, opt_rand, m)
3. (indices, ctr) ← fors_grind(m, R, PK.seed, PK.root, ADRS)
4. sig ← R || toByte(ctr, c)
5. for i from 0 to k - 2:
    a. sk_i ← fors_SKgen(SK.seed, PK.seed, ADRS, i, indices[i])
    b. auth_i ← fors_auth_path(SK.seed, PK.seed, ADRS, i, indices[i])
    c. sig ← sig || sk_i || auth_i
6. return sig
```

9.9 Public Key Recovery

The `fors_pkFromSig` function recovers the FORS+C public key from a signature. For each of the first k-1 trees, it hashes the revealed secret key to obtain the leaf, then iteratively combines with authentication path nodes to compute the tree root. The FORS+C public key is the tweakable hash of these `k-1` roots (tree `k-1` is excluded entirely due to grinding). The function returns $\perp$ if the grinding condition is not satisfied.

```
Algorithm: fors_pkFromSig(sig, m, PK.seed, PK.root, ADRS)
Input:
    sig: FORS+C signature
    m: message
    PK.seed: public seed
    PK.root: public key root
    ADRS: tweak
Output:
    pk: FORS+C public key (n bytes), or $\bot$ if invalid

1. R ← sig[0 : R_SIZE]
2. ctr ← sig[R_SIZE : R_SIZE + c]
3. setTypeAndClear(ADRS, FORS_GRIND)
4. digest ← H_msg_fors(ADRS, R, toByte(ctr, c), PK.seed, PK.root, m)
5. indices ← fors_msg_to_indices(digest)
6. if indices[k-1] != 0:
    return $\bot$
7. roots ← []
8. offset ← R_SIZE + c
9. for i from 0 to k - 2:
    a. sk_i ← sig[offset : offset + n]
    b. auth_i ← sig[offset + n : offset + n + a * n]
    c. offset ← offset + n + a * n
    d. setTypeAndClear(ADRS, FORS_HASH)
    e. setKeyPairAddress(ADRS, i)
    f. setTreeHeight(ADRS, 0)
    g. setTreeIndex(ADRS, indices[i])
    h. node ← Tw(PK.seed, ADRS, sk_i)
    i. idx ← indices[i]
    j. for h from 0 to a - 1:
        setTypeAndClear(ADRS, FORS_TREE)
        setKeyPairAddress(ADRS, i)
        setTreeHeight(ADRS, h + 1)
        setTreeIndex(ADRS, idx >> 1)
        if (idx AND 1) == 0:
            node ← Tw(PK.seed, ADRS, node || auth_i[h * n : (h + 1) * n])
        else:
            node ← Tw(PK.seed, ADRS, auth_i[h * n : (h + 1) * n] || node)
        idx ← idx >> 1
    k. roots[i] ← node
10. setTypeAndClear(ADRS, FORS_PK)
11. pk ← Tw(PK.seed, ADRS, roots[0] || ... || roots[k-2])
12. return pk
```

10 SHRINCS

This section specifies the complete SHRINCS signature scheme by composing the building blocks defined in previous sections.

10.1 Key Generation

The `SHRINCS.KeyGen` function generates a fresh key pair and initializes the signing state. It samples a 3n-byte random seed, partitions it into `SK.seed` (for key derivation), `SK.prf` (for message randomization), and `PK.seed` (for domain separation). It then computes `PK.sf` (the stateful UXMSS root) and `PK.sl` (the stateless hypertree root), combining them into `PK.root` via a tweakable hash. The initial state sets `q=0` with `valid=true`, indicating the stateful path is available starting from signature number 1.

```
Algorithm: SHRINCS.KeyGen()
Output:
    SK: secret key
    PK: public key
    state: initial signing state

1. seed ←$ {0,1}^(3n)
2. SK.seed ← seed[0:n]
3. SK.prf ← seed[n:2n]
4. PK.seed ← seed[2n:3n]
5. ADRS ← new_ADRS()
6. PK.sf ← uxmss_root(SK.seed, PK.seed, ADRS)
7. setLayerAddress(ADRS, d - 1)
8. setTreeAddress(ADRS, 0)
9. PK.sl ← xmss_root(SK.seed, PK.seed, ADRS, h_prime)
10. setTypeAndClear(ADRS, ROOT)
11. PK.root ← Tw(PK.seed, ADRS, PK.sf || PK.sl)
12. SK ← (SK.seed, SK.prf, PK.seed, PK.sf, PK.sl, PK.root)
13. PK ← (PK.seed, PK.root)
14. state ← { q: 0, valid: true }
15. return (SK, PK, state)
```

10.2 Recovery from Seed

The `SHRINCS.Restore` function recovers a key pair from a backed-up seed. This enables the standard Bitcoin workflow of restoring a wallet from a seed phrase. However, since the signing state (which q values have been used) cannot be recovered from the seed alone, the stateful path is disabled by setting `valid=false`. The recovered wallet can only use stateless signatures, which are larger but do not require state tracking. This is the key tradeoff that enables static backups while maintaining security.

```
Algorithm: SHRINCS.Restore(seed)
Input:
    seed: master seed (3n bytes)
Output:
    SK: secret key
    PK: public key
    state: invalid state (stateless only)

1. SK.seed ← seed[0:n]
2. SK.prf ← seed[n:2n]
3. PK.seed ← seed[2n:3n]
4. ADRS ← new_ADRS()
5. PK.sf ← uxmss_root(SK.seed, PK.seed, ADRS)
6. setLayerAddress(ADRS, d - 1)
7. setTreeAddress(ADRS, 0)
8. PK.sl ← xmss_root(SK.seed, PK.seed, ADRS, h')
9. setTypeAndClear(ADRS, ROOT)
10. PK.root ← Tw(PK.seed, ADRS, PK.sf || PK.sl)
11. SK ← (SK.seed, SK.prf, PK.seed, PK.sf, PK.sl, PK.root)
```

```
12. PK ← (PK.seed, PK.root)
13. state ← { q: $\bot$, valid: false }
14. return (SK, PK, state)
```

10.3 Statefull signing

The `SHRINCS.SignStateful` function signs a message using the efficient stateful (UXMSS) path. It first checks that state is valid and that signature slots remain (`q <= h_sf + 1`). It then increments `q` and updates the state before generating the signature. This ordering is critical: if the process is interrupted after signing but before state persistence, the one-time key would be consumed without the state reflecting it, risking catastrophic key reuse on retry (see Section 1.2).

The signature includes `PK.sl` prepended to enable unified verification. This produces minimal signatures (324 bytes for q = 1 in SHRINCS-B) that grow linearly with `q`. If state is invalid or exhausted, the function returns ⊥ and the caller should use stateless signing instead.

```
Algorithm: SHRINCS.SignStateful(m, SK, state)
Input:
    m: message to sign
    SK: secret key (SK.seed, SK.prf, PK.seed, PK.sf, PK.sl, PK.root)
    state: current signing state { q, valid }
Output:
    sig: stateful signature, or $\bot$ if state is invalid/exhausted
    state_prime: updated state

1. if not state.valid:
    return ($\bot$, state)
2. q ← state.q + 1
3. if q > h_sf + 1:
    return ($\bot$, state)
4. state_prime ← { q: q, valid: true }
5. ADRS ← new_ADRS()
6. uxmss_sig ← uxmss_sign(m, SK.seed, SK.prf, PK.seed, PK.root, ADRS, q)
7. sig ← PK.sl || uxmss_sig
8. return (sig, state_prime)
```

10.4 Stateless signing

The `SHRINCS.SignStateless` function signs a message using the stateless (SPHINCS+C) path. This is used when state is lost, corrupted, or exhausted.

SHRINCS uses a unified digest for the stateless path: a single call to `H_msg_fors` produces a digest that encodes both the FORS leaf indices (determining which leaves to reveal in the few-time signature) and the hypertree tree/leaf indices (determining which XMSS trees and WOTS+C key pairs to use at each layer). The grinding counter `ctr` is iterated until the grinding condition is satisfied on this unified digest.

The digest output length is `(roundup(k*a + h_sl) / 8)` bytes. The first `k*a` bits encode the `k` FORS leaf indices (with the last a bits required to be zero by grinding). The remaining `h_sl` bits encode the hypertree path indices.

The signature is constructed bottom-up: first a FORS+C signature on the message, then `d` XMSS signatures on successive layer roots ascending through the hypertree. The signature includes `PK.sf` prepended to enable unified verification.

```
Algorithm: SHRINCS.SignStateless(m, SK)
Input:
    m: message to sign
    SK: secret key
Output:
    sig: stateless signature
```

```
1. ADRS ← new_ADRS()
2. opt_rand ← random(n) or PK.seed
3. R ← PRF_msg(SK.prf, PK.seed, opt_rand, m)
4. setTypeAndClear(ADRS, FORS_GRIND)
5. for ctr from 0 to 2^32 - 1:
    a. digest ← H_msg_fors(ADRS, R, ctr, PK.seed, PK.root, m)
    b. indices ← fors_msg_to_indices(digest)
    c. if indices[k-1] == 0:
        break
6. if ctr == 2^32:
    abort with error
7. (tree_idx, leaf_idx) ← parse_idx(digest)
8. setLayerAddress(ADRS, 0)
9. setTreeAddress(ADRS, tree_idx[0])
10. fors_sig ← R || ctr
11. for i from 0 to k - 2:
    a. sk_i ← fors_SKgen(SK.seed, PK.seed, ADRS, i, indices[i])
    b. auth_i ← fors_auth_path(SK.seed, PK.seed, ADRS, i, indices[i])
    c. fors_sig ← fors_sig || sk_i || auth_i
12. fors_pk ← fors_pkFromSig(fors_sig, m, PK.seed, PK.root, ADRS)
13. ht_sig ← []
14. msg ← fors_pk
15. for layer from 0 to d - 1:
    a. setLayerAddress(ADRS, layer)
    b. setTreeAddress(ADRS, tree_idx[layer])
    c. xmss_sig ← xmss_sign(msg, SK.seed, SK.prf, PK.seed, PK.root, ADRS, h', leaf_idx[layer])
    d. ht_sig ← ht_sig || xmss_sig
    e. if layer < d - 1:
        msg ← xmss_root(SK.seed, PK.seed, ADRS, h')

16. sig ← PK.sf || fors_sig || ht_sig
17. return sig
```

10.5 Index Parsing

The `parse_idx` function extracts tree and leaf indices for each hypertree layer from the stateless digest. In the unified digest scheme, these bits follow the `k*a` bits used for FORS indices. The `h_sl` bits are partitioned into `d` pairs of `(leaf_idx, tree_idx)`, extracted from least significant (layer 0) to most significant (layer `d-1`), with `h'` bits per leaf index.

```
Algorithm: parse_idx(digest)
Input:
    digest: message digest
Output:
    tree_idx: array of d tree indices
    leaf_idx: array of d leaf indices

1. Extract hypertree bits starting after FORS indices
2. ht_offset ← k * a
3. ht_bits ← extract_bits(digest, ht_offset, h_sl)
4. idx ← ht_bits
5. tree_idx ← []
6. leaf_idx ← []
7. for layer from 0 to d - 1:
    a. leaf_idx[layer] ← idx AND (2^h' - 1)
```

```
        b. idx ← idx >> h'
        c. tree_idx[layer] ← idx AND (2^(h_sl - (layer + 1) * h') - 1)
    8. return (tree_idx, leaf_idx)
```

Note: The total digest must encode `k*a + h_sl = 6*22 + 40 = 172` bits. The `H_msg_fors` output length (Section 5.4.2) must be set accordingly: `roundup((k*a + h_sl) / 8) = 22` bytes. This is a correction from the original `roundup((k*a) / 8) = 17` bytes, which was insufficient for the unified digest.

10.6 Statefull verification

The `SHRINCS.VerifyStateful` function verifies a stateful signature. It extracts `PK.sl` from the signature, parses the UXMSS signature, and infers `q` from the authentication path length. It then uses the two-phase WOTS+C hashing to recover `PK.sf` via `uxmss_pkFromSig` and verifies that `Tw(PK.seed, ADRS, PK.sf || PK.sl)` equals `PK.root`.

```
Algorithm: SHRINCS.VerifyStateful(m, sig, PK)
Input:
    m: message
    sig: stateful signature
    PK: public key (PK.seed, PK.root)
Output:
    valid: boolean

1. ADRS ← new_ADRS()
2. PK.sl ← sig[0 : n]
3. uxmss_sig ← sig[n : ]
4. wots_sig_size ← R_SIZE + c + l * n
5. auth_len ← len(uxmss_sig) - wots_sig_size
6. if auth_len < 0 or (auth_len mod n) != 0:
    return false
7. q ← auth_len / n
8. if q < 1 or q > h_sf + 1:
    return false
9. wots_sig ← uxmss_sig[0 : wots_sig_size]
10. auth ← uxmss_sig[wots_sig_size : wots_sig_size + q * n]
11. PK.sf ← uxmss_pkFromSig(wots_sig, auth, m, PK.seed, PK.root, ADRS, q)
12. if PK.sf == $\bot$:
    return false
13. setTypeAndClear(ADRS, ROOT)
14. expected_root ← Tw(PK.seed, ADRS, PK.sf || PK.sl)
15. return (expected_root == PK.root)
```

10.7 Stateless verification

The `SHRINCS.VerifyStateless` function verifies a stateless signature. It extracts `PK.sf` from the signature, recomputes the unified digest from the FORS+C signature's `R` and `ctr`, extracts both FORS indices and hypertree indices from the same digest, then verifies bottom-up through the hypertree.

```
Algorithm: SHRINCS.VerifyStateless(m, sig, PK)
Input:
    m: message
    sig: stateless signature
    PK: public key (PK.seed, PK.root)
Output:
    valid: boolean
```

```
1. ADRS ← new_ADRS()
2. PK.sf ← sig[0 : n]
3. offset ← n
4. fors_sig_size ← R_SIZE + c + (k - 1) * (n + a * n)
5. fors_sig ← sig[offset : offset + fors_sig_size]
6. offset ← offset + fors_sig_size
7. R ← fors_sig[0 : R_SIZE]
8. ctr ← fors_sig[R_SIZE : R_SIZE + c]
9. setTypeAndClear(ADRS, FORS_GRIND)
10. digest ← H_msg_fors(ADRS, R, ctr, PK.seed, PK.root, m)
11. indices ← fors_msg_to_indices(digest)
12. if indices[k-1] != 0:
    return false
13. (tree_idx, leaf_idx) ← parse_idx(digest)
14. setLayerAddress(ADRS, 0)
15. setTreeAddress(ADRS, tree_idx[0])
16. fors_pk ← fors_pkFromSig(fors_sig, m, PK.seed, PK.root, ADRS)
17. if fors_pk == $\bot$:
    return false
18. msg ← fors_pk
19. xmss_sig_size ← R_SIZE + c + l * n + h' * n
20. for layer from 0 to d - 1:
    a. xmss_sig ← sig[offset : offset + xmss_sig_size]
    b. offset ← offset + xmss_sig_size
    c. wots_sig ← xmss_sig[0 : R_SIZE + c + l * n]
    d. auth ← xmss_sig[R_SIZE + c + l * n : R_SIZE + c + l * n + h' * n]
    e. setLayerAddress(ADRS, layer)
    f. setTreeAddress(ADRS, tree_idx[layer])
    g. msg ← xmss_pkFromSig(wots_sig, auth, msg, PK.seed, PK.root, ADRS, h', leaf_idx[layer])
    h. if msg == $\bot$:
        return false
21. PK.sl ← msg
22. setTypeAndClear(ADRS, ROOT)
23. expected_root ← Tw(PK.seed, ADRS, PK.sf || PK.sl)
24. return (expected_root == PK.root)
```

10.8 Unified Verification

The `SHRINCS.Verify` function provides a single entry point for verifying either stateful or stateless signatures. It distinguishes between the two types based on signature length: stateful signatures have a maximum size of `max_sf_size = n + R_SIZE + c + l*n + (h_sf + 1)*n` bytes, while stateless signatures are strictly larger (see Section 8.7 for the derivation that this bound always holds).

```
Algorithm: SHRINCS.Verify(m, sig, PK)
Input:
    m: message
    sig: signature (stateful or stateless)
    PK: public key (PK.seed, PK.root)
Output:
    valid: boolean

1. max_sf_size ← n + R_SIZE + c + l * n + (h_sf + 1) * n
2. if len(sig) <= max_sf_size:
        return SHRINCS.VerifyStateful(m, sig, PK)
    else:
        return SHRINCS.VerifyStateless(m, sig, PK)
```

11 Data structures

This section specifies the binary encoding of SHRINCS keys and signatures. All multi-byte integers are encoded in big-endian byte order unless otherwise specified.

11.1 Secret Key Format

The secret key consists of six `n`-byte values:

```
SK.seed     : n bytes   // Seed for key derivation
SK.prf      : n bytes   // Secret for PRF_msg
PK.seed     : n bytes   // Domain separator
PK.sf       : n bytes   // Stateful public key
PK.sl       : n bytes   // Stateless public key
PK.root     : n bytes   // Combined public key
```

Total size: `6n = 96 bytes`. The first `3n` bytes (`SK.seed || SK.prf || PK.seed`) constitute the master seed that can be backed up using standard seed phrase mechanisms. The remaining `3n` bytes can be deterministically recomputed from the seed.

11.2 Public Key Format

The public key consists of two `n`-byte values:

```
PK.seed     : n bytes   // Domain separator
PK.root     : n bytes   // Root public key
```

Total size: `2n = 32 bytes`.

11.3 Stateful Signature Format

A stateful signature consists of the stateless public key followed by a UXMSS signature:

```
PK.sl       : n bytes          // Stateless public key
R           : R_SIZE bytes     // Randomness for message hashing
ctr         : c bytes          // Grinding counter
chain_values: l * n bytes      // WOTS+C signature
auth_path   : q * n bytes      // UXMSS authentication path
```

The signature size depends on the signature index `q`:

```
SHRINCS-B (n=16, R_SIZE=32, c=4, l=16):
    PK.sl       : 16 bytes
    R           : 32 bytes
    ctr         : 4 bytes
    chain_values: 256 bytes
    auth_path   : q * 16 bytes

    Total: 308 + 16*q bytes

SHRINCS-L (n=16, R_SIZE=32, c=4, l=64):
    PK.sl       : 16 bytes
    R           : 32 bytes
    ctr         : 4 bytes
    chain_values: 1024 bytes
    auth_path   : q * 16 bytes

    Total: 1076 + 16*q bytes
```

11.4 Stateless Signature Format

A stateless signature consists of the stateful public key, a FORS+C signature, and `d` XMSS signatures forming the hypertree:

```
PK.sf       : n bytes                       // Stateful public key
FORS+C signature:
    fors_R      : R_SIZE bytes              // Randomness
    fors_ctr    : c bytes                   // Grinding counter
    fors_sigs   : (k-1) * (n + a*n) bytes   // k-1 trees
Hypertree: d XMSS signatures. For each layer i:
    xmss_R[i]     : R_SIZE bytes            // Randomness for WOTS+C
    xmss_ctr[i]   : c bytes                 // Grinding counter for WOTS+C
    xmss_chains[i]: l * n bytes             // WOTS+C chain values
    xmss_auth[i]  : h' * n bytes            // XMSS authentication path

SHRINCS-B (n=16, R_SIZE=32, c=4, k=6, a=22, d=5, h'=8, l=16):
    PK.sf       : 16 bytes
    FORS+C signature:
        fors_R    : 32 bytes
        fors_ctr  : 4 bytes
        fors_sigs : 5 * (16 + 22*16) = 5 * 368 = 1840 bytes
        subtotal: 1892 bytes
    XMSS hypertree (5 layers):
        per layer:
            R           : 32 bytes
            ctr         : 4 bytes
            chain_values: 16 * 16 = 256 bytes
            auth_path   : 8 * 16 = 128 bytes
            per layer subtotal: 420 bytes
        5 layers: 5 * 420 = 2100 bytes

    Total: 16 + 1892 + 2100 = 4008 bytes

SHRINCS-L (n=16, R_SIZE=32, c=4, k=6, a=22, d=5, h'=8, l=64):
    PK.sf       : 16 bytes
    FORS+C signature:
        fors_R    : 32 bytes
        fors_ctr  : 4 bytes
        fors_sigs : 5 * (16 + 22*16) = 1840 bytes
        subtotal: 1892 bytes
    XMSS hypertree (5 layers):
        per layer:
            R           : 32 bytes
            ctr         : 4 bytes
            chain_values: 64 * 16 = 1024 bytes
            auth_path   : 8 * 16 = 128 bytes
            per layer subtotal: 1188 bytes
        5 layers: 5 * 1188 = 5940 bytes

    Total: 16 + 1892 + 5940 = 7848 bytes
```

11.5 Signature Type Indication

Verifiers distinguish between stateful and stateless signatures by length (Algorithm 10.8):

```
max_stateful_size = n + R_SIZE + c + l*n + (h_sf + 1)*n
```

- For SHRINCS-B: `max_stateful_size = 16 + 32 + 4 + 256 + 227*16 = 3940` bytes
- For SHRINCS-L: `max_stateful_size = 16 + 32 + 4 + 1024 + 419*16 = 7780` bytes

Signatures with length `<= max_stateful_size` are processed as stateful; larger signatures as stateless.